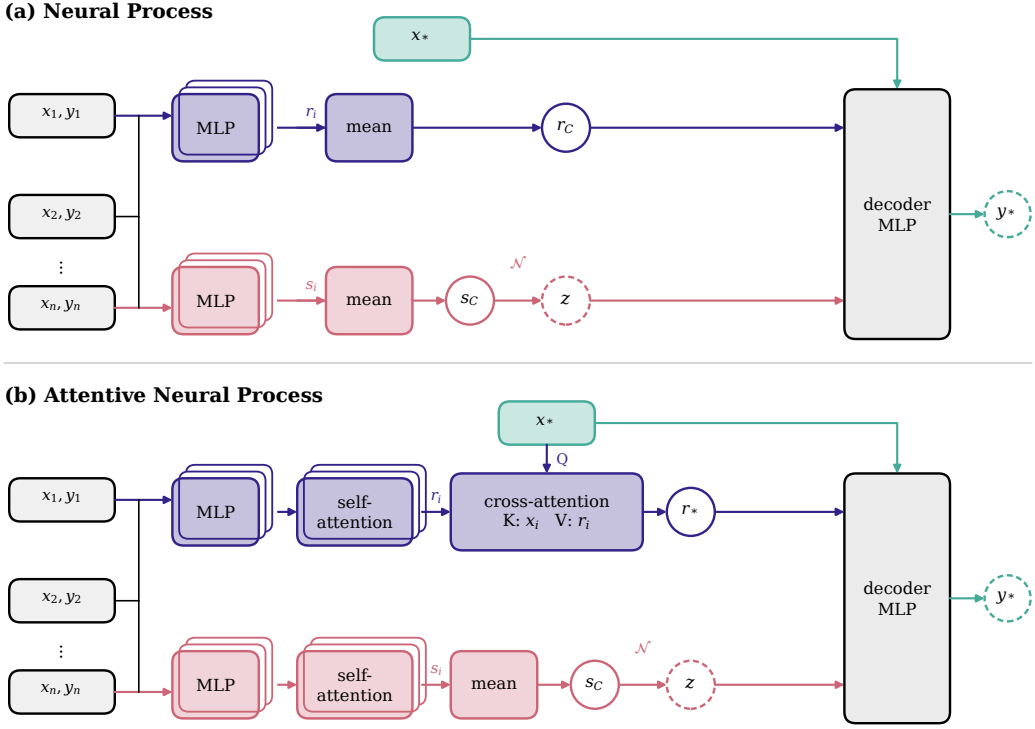
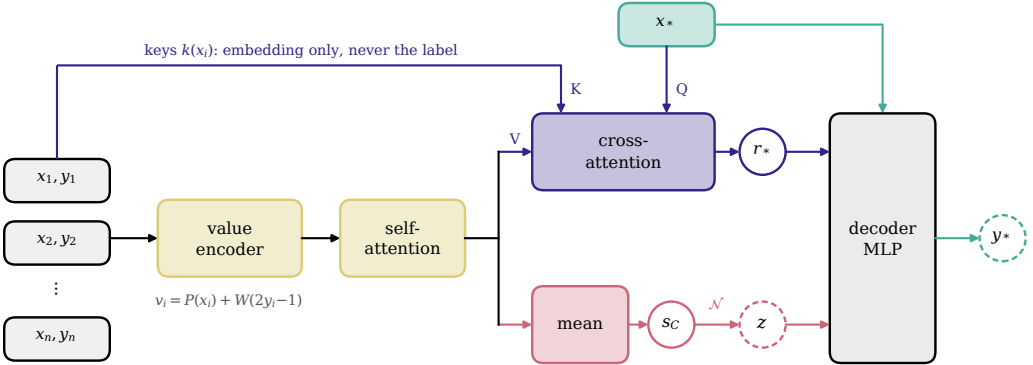
Neural Process vs Attentive Neural Process

After Kim et al. (2019), Fig. 2. Blue: deterministic. Red: latent. Teal: target. Stacked boxes: applied to every context pair.

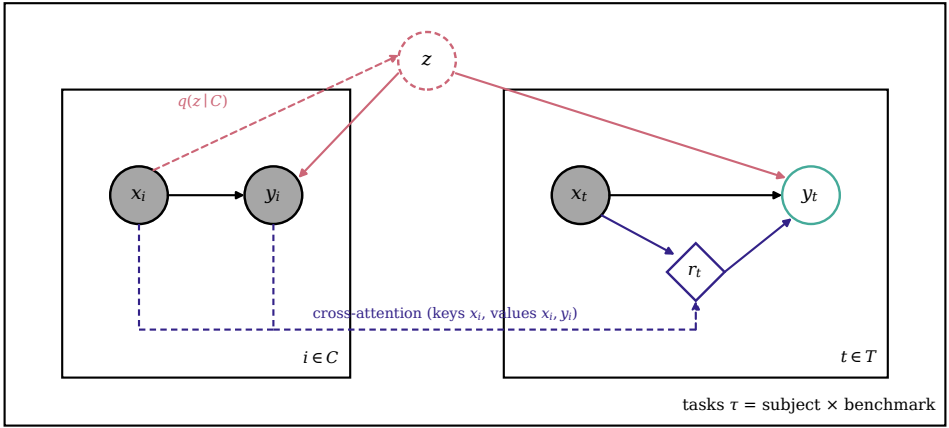


Our Attentive Neural Process

Left: context C. Sand: shared by both paths. Blue: deterministic. Red: latent. Teal: target. $d = 64$, 4 heads, $z \in \mathbb{R}^{32}$.



Train: $z \sim q(z \mid C \cup T)$; loss BCE + KL($q(z \mid C \cup T) \,\|\, q(z \mid C)$). Prior-anchor variant: + Beta-Binomial logit of y_C.

Graphical model of the ANP

Shaded: observed. Dashed circle: latent. Diamond: deterministic. Solid edge: generative. Dashed edge: inference $q(z \mid C)$.